

SignNav: Directional Signage-Cue Guided Visual Navigation in Large-Scale Indoor Environments

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Abstract—Humans routinely leverage directional cues from signage to navigate toward destinations in unfamiliar Large-Scale Indoor (LSI) environments, such as hospitals and airport terminals. However, learning how embodied agents can use sparse and target-relevant signage cues for closed-loop navigation remains underexplored. This paper studies SignNav, a directional signage-cue guided visual navigation task in which an agent navigates to a semantic target using egocentric observations together with directional cues and cue locations provided by an upstream sign parser. This formulation focuses on downstream cue-conditioned policy learning, rather than open-ended signage parsing, OCR, or compound sign reasoning. To facilitate research in this domain, we construct the LSI-Dataset, a large-scale benchmark with signage cues-labeled indoor environments and automatically generated trajectory-action episodes. SignNav presents two key challenges: grounding transient directional cues into the current observation, and maintaining temporal memory when signage is sparse or temporarily unavailable. To address these challenges, we propose the Spatial-Temporal Aware Transformer (START), which combines a spatial-aware module for cue-observation grounding with a temporal-aware module for history-based decision making. Extensive experiments, including cue-representation ablations, sparse-signage evaluations, and real-world robot trials, demonstrate the effectiveness of START for directional signage-cue guided navigation in large-scale indoor environments.

Index Terms—Autonomous agents, perception-action coupling, signage-driven navigation, vision-based navigation.

I. INTRODUCTION

NAVIGATING Large-Scale Indoor (LSI) environments, such as airport terminals and hospitals, remains a fundamental challenge for embodied agents. In such spaces, target

Received 16 March 2026; accepted 4 August 2026. Date of publication 2 September 2026; date of current version 14 September 2026. This article was recommended for publication by Associate Editor J. S. B. Perez and Editor L. Jamone upon evaluation of the reviewers' comments. This work was supported by the Fundo para o Desenvolvimento das Ciências e da Tecnologia (FDCT) of Macao SAR under Grant 0067/2023/AFJ and Grant 0117/2024/RIB2. (Corresponding author: Hui Kong.)

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This article has supplementary downloadable material available at <https://doi.org/10.1109/LRA.2026.3730302>, provided by the authors.

Digital Object Identifier 10.1109/LRA.2026.3730302

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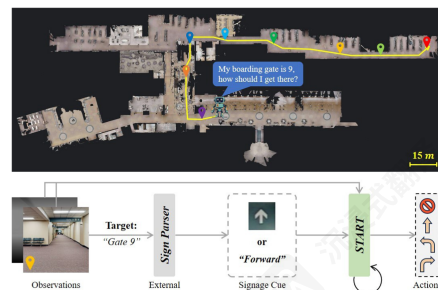


Fig. 1. Directional signage-cue guided navigation in a LSI environment. Humans use signage to extract target-relevant directional cues when navigating unfamiliar public spaces. SignNav studies the downstream navigation problem in which an agent uses parsed directional cues from signage, together with egocentric observations and history, to reach the target location. The directional signage-cue can be represented as either visual crop image or structured category.



Fig. 2. An episode example in a hospital scene. The agent is required to make action decisions according to the directional signage-cue and finally arrive at the target location. Different scenes have different styles of signage.

locations are often not directly visible from the start, and successful navigation requires the agent to exploit intermediate guidance provided by the environment. Humans naturally use signage for this purpose: rather than following a pre-given step-by-step instruction, they usually observe local signs, extract target-relevant directional cues, and use these cues to decide how to proceed (see Fig. 1). Enabling robots to use such environment-embedded guidance is therefore an important step toward practical indoor navigation.

Existing embodied navigation paradigms [1] have made substantial progress but do not fully capture this setting. Object Navigation (ObjectNav) [2] typically requires the agent to search for an object category and assumes that the goal can eventually be visually recognized, while Vision-and-Language Navigation (VLN) [3] relies on a complete natural-language instruction given before navigation starts. Recent methods based on large

SignNav: 大型室内环境中的方向性标志 - 线索引导视觉导航

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摘要——人类通常利用方向性线索从标志中导航, 以在大型室内 (LSI) 环境中前往目的地, 例如医院和机场航站楼。然而, 关于具身代理如何利用稀疏且与目标相关的标志线索进行闭环导航的研究仍较少。本文研究了 SignNav, 这是一个由方向性标志线索引导的视觉导航任务, 其中代理利用自眼观测、上游标志解析器提供的方向性线索及线索位置, 导航至语义目标。该任务侧重于下游基于线索的条件策略学习, 而非开放式的标志解析、OCR 或复合标志推理。为了促进该领域的研究, 我们构建了 LSI-Dataset, 这是一个包含带标志线索标注的室内环境的大规模基准, 并自动生成轨迹 - 动作片段。SignNav 提出了两个关键挑战: 将瞬态方向性线索锚定到当前观测中, 以及在标志稀疏或暂时不可用时保持时间记忆。为应对这些挑战, 我们提出了时空感知 Transformer (START), 它将一个用于线索 - 观测锚定的空间感知模块与一个用于基于历史决策的时间感知模块相结合。广泛的实验, 包括线索表示消融、稀疏标志评估以及真实世界机器人试验, 证明了 START 在大型室内环境中进行方向性标志线索引导导航的有效性。

索引词——自主代理, 感知 - 行动耦合, 标志驱动的导航, 基于视觉的导航。

AVIGATING 一、引言

NAVIGATING 大规模室内 (LSI) 环境, 如机场航站楼和医院, 仍是具身智能体面临的一项基本挑战。在这样的空间中, 目标

2026 年 3 月 16 日收到; 2026 年 8 月 4 日接受。出版日期 2026 年 9 月 2 日。本文推荐发表, 并在评审人意见评估后。此项工作

由澳门特别行政区科学和技术发展基金 (FDCT) 资助, 项目编号 0067/2023/AFJ 和 0117/2024/RIB2。 (通讯作者: 孔慧。)

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<https://doi.org/10.1109/LRA.2026.3730302>。

数字对象标识符 10.1109/LRA.2026.3730302

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图 1. 在 LSI 环境中基于方向性标志线索的导航。人类在导航不熟悉公共空间时, 利用标志提取与目标相关的方向性线索。SignNav 研究下游导航问题, 其中智能体利用从标志解析出的方向性线索, 结合自观测和历史信息, 到达目标位置。方向性标志线索可表示为视觉裁剪图像或结构化类别。

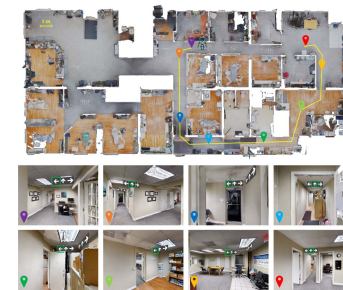


图 2. 医院场景中的一个示例。智能体需要根据方向标志提示做出行动决策, 并最终到达目标位置。不同场景的标志风格各不相同。

位置通常无法直接从起点直接看到, 成功的导航需要智能体利用环境提供的中间引导。人类自然利用路标来实现这一目的: 他们通常不会遵循预先给定的逐步指令, 而是观察当地路标, 提取与目标相关的方向线索, 并利用这些线索决定如何继续前进 (见图 1)。因此, 使机器人能够使用这种嵌入环境中的引导, 是实现实用室内导航的重要一步。

现有的具身导航范式 [1] 已取得实质性进展, 但尚未完全涵盖该场景。物体导航 (ObjectNav) [2] 通常要求代理搜索物体类别, 并假设目标最终可以被视觉识别, 而视觉 - 语言导航 (VLN) [3] 则依赖于导航开始前提供的完整自然语言指令。基于大型

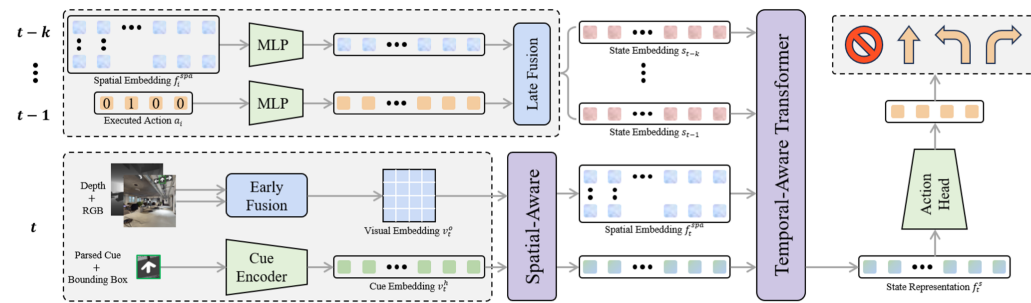


Fig. 3. The architecture of our Spatial-Temporal Aware Transformer (START) model for the SignNav task. START uses a spatial-aware module to ground the parsed cue and its location into the current observation, and a temporal-aware module to integrate historical states when cues are sparse or temporarily unavailable. The parsed cue can be represented as either visual crop image or structured category, such as “forward” and “right”.

language or vision-language models improve open-vocabulary reasoning [4], [5], [6], [7], but they are often evaluated in relatively small household environments or depend on modular pipelines with explicit mapping and planning. In contrast, signage-guided navigation in LSI environments requires the agent to react to sparse and dynamically observed cues that become available only when the agent reaches the corresponding regions.

In this work, we study **SignNav**, a directional signage-cue guided visual navigation task for large-scale indoor environments. To clarify the scope of the problem, we separate sign perception from downstream navigation. Given a target g , an upstream sign parser extracts the target-relevant directional cue and its location from the current observation. The navigation policy then uses this parsed cue, together with egocentric observations and navigation history, to predict the next action. This formulation focuses on how an embodied agent grounds directional signage cues into physical navigation behavior, rather than claiming to solve open-ended signage parsing, compound sign reasoning, or general visual sign grounding.

To support this task, we construct the **LSI-Dataset**, which contains signage-labeled LSI environments and automatically generated trajectory-action episodes. The dataset is built through a procedural signage injection and annotation pipeline, allowing directional cues and their spatial locations to be recorded during navigation. SignNav introduces two key challenges for robots. (1) **Dynamic Cue Grounding**: The same cue can imply different actions depending on where it appears in the egocentric scene and what navigable structure is currently visible. (2) **Sparse Observability**: Signage is sparse and may be observed before the corresponding junction becomes actionable, requiring temporal memory to retain previously observed cues and use them later. These challenges distinguish **SignNav** from settings where the instruction is fixed at the beginning of the episode or where the goal object is directly searched for in the scene.

To address these challenges, we propose the **Spatial-Temporal Aware Transformer (START)**, a closed-loop policy for directional signage-cue guided navigation (Fig. 3). START uses a spatial-aware module to ground the parsed cue and its location into the current egocentric observation, and a temporal-aware module to integrate historical states when cues are sparse or temporarily unavailable. We evaluate START against representative rule-based, VLM-based, and learning-based baselines, and

further analyze cue representations, bounding-box information, signage sparsity, and real-world deployment. The results show that START effectively exploits directional signage cues for navigation in large-scale indoor environments, while also revealing important limitations and future directions for richer signage understanding and more complex public-space behaviors.

II. RELATED WORK

A. Object Goal Navigation

Object Goal Navigation (ObjectNav) [2] requires an agent to find an instance of a specified object category in an unseen environment. Existing methods commonly rely on reinforcement learning [8], [9], imitation learning [10], semantic mapping [11], [12], or open-vocabulary visual representations [4], [5], [13], [14] to improve search efficiency. However, ObjectNav is mainly designed for targets that can eventually be visually recognized. In LSI environments, destinations such as boarding gates or service areas may not be directly observable from a distance, and navigation often depends on intermediate directional information provided by signage. Our work therefore focuses on using target-relevant directional cues encountered during navigation rather than directly searching for the final target object.

B. Vision-and-Language Navigation

Vision-and-Language Navigation (VLN) studies how agents follow natural-language instructions to reach a goal [3], [15]. Recent extensions use pre-trained vision-language models or large language models to improve instruction following and generalization [7], [16], [17], [18]. Signage-guided navigation is closely related to VLN because signs can be regarded as environment-embedded instructions. The difference is that signage cues are local, sparse, and observed online during execution, rather than being provided as a complete global instruction at the beginning of an episode. SignNav studies this setting by conditioning navigation on target-relevant directional cues extracted from visible signage.

C. Signage-Aware and Cue-Conditioned Navigation

Signage has been studied in robotics and computer vision through sign detection [19], [20], text parsing [21], [22], visual sign grounding [23], [24], and mapless navigation [25]. Recent

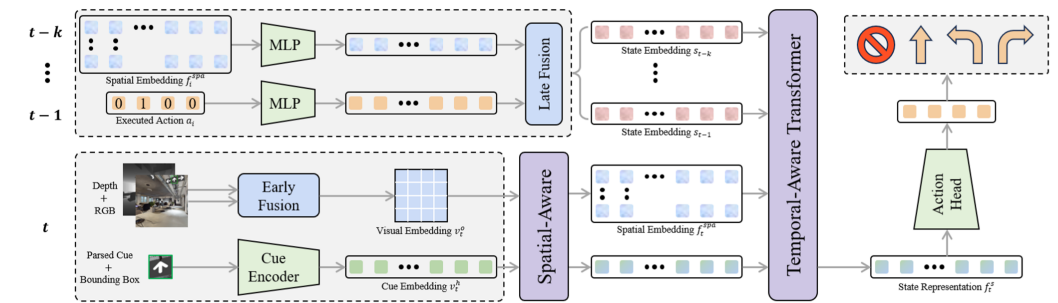


图 3. 我们用于 SignNav 任务的时空感知 Transformer (START) 模型架构。START 使用空间感知模块将解析的线索及其位置定位到当前观测中，并使用时间感知模块在线索稀疏或暂时不可用时整合历史状态。解析的线索可以表示为视觉裁剪图像或结构化类别，例如“前进”和“右转”。

语言或视觉 - 语言模型提升了开放词汇推理能力

[4], [5], [6], [7], 但通常在小规模家庭环境中进行评估, 或依赖于具有显式映射和规划的模块化管道。相比之下, 在 LSI 环境中, 标志引导的导航要求代理对稀疏且动态观察到的线索做出反应, 而这些线索仅在代理到达相应区域时才变得可用。

在本工作中, 我们研究 SignNav, 这是一个面向大规模室内环境的、由方向性标志线索引导的视觉导航任务。为明确问题范围, 我们将标志感知与下游导航任务分离。给定目标 g , 上游标志解析器从当前观测中提取与目标相关的方向性线索及其位置。导航策略随后利用该解析出的线索, 结合第一人称观测和导航历史, 预测下一步动作。该框架聚焦于具身智能体如何将方向性标志线索落地为物理导航行为, 而非声称解决开放式的标志解析、复合标志推理或通用视觉标志对齐问题。

为了支持此任务, 我们构建了 LSI-Dataset, 该数据集包含带有路牌标注的 LSI 环境以及自动生成的轨迹 - 动作片段。该数据集通过程序化的路牌注入与标注流程构建, 使得在导航过程中能够记录方向提示及其空间位置。SignNav 为机器人引入了两个关键挑战: (1) 动态提示 grounding: 同一提示在自中心视图中出现的位置不同, 或当前可见的可通行结构不同时, 可能暗示不同的动作。(2) 稀疏可观测性: 路牌稀疏, 可能在对应路口变得可操作之前就被观察到, 因此需要时间记忆来保留先前观察到的提示, 并在后续使用时。这些挑战使 SignNav 区别于指令在片段开始时即固定, 或直接在场景中搜索目标物体的设置。

为解决这些挑战, 我们提出了时空感知 Transformer (START), 这是一种用于方向标志线索引导导航的闭环策略 (图 3)。START 使用空间感知模块将解析出的线索及其位置锚定到当前的自我中心观测中, 并使用时间感知模块在线索稀疏或暂时不可用时整合历史状态。我们将 START 与代表性的基于规则、基于视觉语言模型 (VLM) 和基于学习的基线方法进行了评估, 并

进一步分析线索表示、边界框信息、标志牌稀疏性以及现实世界部署。结果表明, START 能有效利用方向性标志线索在大型室内环境中进行导航, 同时也揭示了更丰富的标志理解及更复杂的公共空间行为方面的重要局限性与未来方向。

II. 相关工作

A. 目标导航

目标导向导航 (ObjectNav) [2] 需要智能体在未见过的环境中定位指定类别的某个实例。现有方法通常依赖强化学习 [8], [9], 模仿学习 [10], 语义映射 [11], [12], 或开放词汇视觉表征 [4], [5], [13], [14] 来提高搜索效率。然而, ObjectNav 主要针对最终可被视觉识别的目标进行设计。在 LSI 环境中, 如登机口或服务区域等目的地可能无法从远处直接观察, 导航往往依赖于标识牌提供的中间方向信息。因此, 我们的工作侧重于利用导航过程中遇到的与目标相关的方向线索, 而非直接搜索最终目标物体。

B. 视觉 - 语言导航

视觉 - 语言导航 (VLN) 研究智能体如何遵循自然语言指令到达目标 [3], [15]。最近的扩展工作利用预训练的视觉 - 语言模型或大型语言模型来改进指令遵循能力和泛化能力 [7], [16], [17], [18]。标志引导导航与 VLN 密切相关, 因为标志可被视为嵌入环境中的指令。不同之处在于, 标志线索是局部的、稀疏的, 并在执行过程中在线观测, 而非在回合开始时以完整的全球指令形式提供。SignNav 通过针对从可见标志中提取的目标相关方向线索来研究这一设置。

C. 标志感知与线索条件导航

标志在机器人学和计算机视觉中已通过标志检测 [19], [20], 文本解析 [21], [22], 视觉标志定位 [23], [24], 以及无地图导航 [25] 进行了研究。最近

works such as ReasonNav [26] and SignScene [27] further highlight the importance and difficulty of interpreting navigational signs in real environments, including the grounding of symbolic sign information to local spatial structure. Similarly, concurrent works [28], [29] explore signage-based navigation but rely on modular pipelines that pair LLMs with traditional planners. Other cue- or command-conditioned navigation methods, such as DECISION [30] and Intention-Net [31], learn navigation behaviors from discrete intentions or local goal cues. These works are highly relevant to our setting, but they either address broader sign understanding and modular reasoning, or assume that high-level commands are already available for low-level control. In contrast, our work focuses on downstream directional-cue guided visual navigation: given a parsed target-relevant cue and its location, the agent learns to ground this information in egocentric observations and leverage temporal context for predicting closed-loop navigation actions.

III. DIRECTIONAL SIGNAGE-CUE GUIDED VISUAL NAVIGATION

This work focuses on downstream navigation conditioned on signage-derived directional cues in large-scale indoor environments. We decouple raw signage parsing from policy learning: given the current observation and target, an upstream parser (such as [22]) provides the target-relevant directional cue and its image location, while the navigation policy grounds this cue in the egocentric observation and incorporates temporal context to predict actions. This formulation isolates directional-cue guided navigation from open-ended OCR, compound sign reasoning, and general signage understanding. In simulation, the parser is implemented by procedural annotations; in real-world deployment, it can be instantiated by a vision-language sign parser.

A. Task Definition

We formulate **SignNav** as a Partially Observable Markov Decision Process (POMDP). The objective is for an embodied agent to navigate in a continuous 3D environment toward a semantic target g by using target-relevant directional cues observed from signage.

Observation Space: At each time step t , the agent receives an egocentric RGB-D observation $o_t = \{o_t^r, o_t^d\}$. An upstream sign parser extracts the directional cue and its image location as:

$$\psi(o_t, g) \rightarrow (c_t, b_t) \quad (1)$$

where c_t denotes the parsed target-relevant cue and $b_t = [x_{min}, y_{min}, x_{max}, y_{max}]$ denotes its bounding box. The cue c_t can be represented as either a visual crop or a structured direction category, depending on the parser interface.

Action Space: The agent uses a discrete action space $a_t \in \mathcal{A}$ consisting of four low-level commands: Move Forward (0.25m), Turn Left (15°), Turn Right (15°), and Stop. This setup follows common continuous-environment navigation protocols [15].

Policy Learning: Since signage cues are sparse and may be observed before the corresponding junction becomes actionable, the agent must use historical context. We define the history as:

$$\mathcal{H}_t \triangleq \{(o_0, c_0, b_0, a_0), \dots, (o_{t-1}, c_{t-1}, b_{t-1}, a_{t-1})\} \quad (2)$$

The goal is to learn a policy π , parameterized by Θ , that predicts the next navigation action according to:

$$a_t \sim \pi(a_t | \mathcal{H}_t, o_t, c_t, b_t; \Theta) \quad (3)$$

The episode terminates when the agent issues Stop near the target or reaches the maximum step limit.

B. LSI-Dataset Creation

Existing embodied navigation datasets [32], [33] mainly focus on household-scale environments and do not fully capture the spatial scale and topology of public indoor spaces. To support **SignNav**, we construct the **Large-Scale Indoor (LSI) Dataset** from 20 high-fidelity 3D scans collected from open-source repositories [32], [34], [35], including hospitals, offices, and other large indoor environments. These scenes contain long corridors, open halls, and complex topological structures, making them suitable for evaluating long-horizon signage-cue guided navigation.

Signage Injection: We develop an automated Blender-based pipeline to procedurally inject signage into the 3D meshes. Candidate sign locations are selected around decision points such as intersections and corners, and directional cues are assigned according to the local route geometry. To reduce overfitting to a single visual template, the dataset includes diverse signage appearances, including variations in arrow shape, color, layout, and background style. Fig. 2 illustrates the rendered effect in a hospital scene. In this work, we focus on directional cues and leave richer signage understanding, such as OCR and compound instruction parsing, to upstream perception modules.

Cue Representation: For each visible target-relevant sign, the dataset provides the directional cue in two complementary forms: (1) a visual cue, represented by the cropped image region of the directional symbol, and (2) a structured cue, represented by the parsed direction category, e.g., *left*, or *forward*. Both cue forms are paired with the same image-space bounding box b_t , allowing models to study either visual cue grounding or parser-based structured cue grounding under a unified navigation protocol.

Episode Generation: For each scene, we construct a navigable connectivity graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ using Poisson Disk Sampling [36]. Start-goal pairs are sampled from graph nodes $\mathcal{V}$, and shortest paths are planned using Dijkstra's algorithm followed by spline-based smoothing. An oracle agent then executes the smoothed trajectory to generate training and evaluation episodes. To automatically annotate the directional signage cues for each episode, we determine the oracle cue direction by leveraging the local curvature of the smoothed trajectory and the agent's heading relative to the injected signage. At each time step t , we record the RGB-D observation o_t , the target-relevant directional cue c_t , its bounding box b_t , and the oracle action a_t^* :

$$\tau = \{(o_t, c_t, b_t, a_t^*)\}_{t=0}^T \quad (4)$$

where T is the episode horizon. In simulation, c_t and b_t are obtained from procedural annotations, providing a controlled interface for studying downstream directional-cue guided navigation.

诸如 ReasonNav [26] 和 SignScene [27] 等工作进一步凸了解释真实环境中导航标志的重要性与难度, 包括将符号标志信息锚定到局部空间结构。同样, 并发工作

[28], [29] 探索基于标志的导航, 但依赖将大语言模型与传统规划器配对的分段式流水线。其他基于线索或命令条件的导航方法, 如 DECISION [30] 和 Intention-Net [31], 从离散意图或局部目标线索中学习导航行为。这些工作与我们的设置高度相关, 但它们要么关注更广泛的标志理解和模块化推理, 要么假设高层命令已可用于底层控制。相比之下, 我们的工作专注于下游基于方向线索的视觉导航: 给定解析出的与目标相关的线索及其位置, 智能体学习将该信息锚定到自观测中, 并利用时间上下文来预测闭环导航动作。

三. 方向标志 - 提示引导视觉导航

本工作关注于基于大规模室内环境中从标识牌提取的方向性线索进行下游导航。我们将原始标识牌解析与策略学习解耦: 给定当前观测和目标, 上游解析器(如 [22])提供与目标相关的方向性线索及其图像位置, 而导航策略将该线索与第一人称观测相结合, 并融入时间上下文以预测动作。该框架将方向性线索引导的导航与开放式 OCR、复合标识牌推理及通用标识牌理解分离开来。在仿真中, 解析器通过过程化注释实现; 在实际部署中, 它可由视觉 - 语言标识牌解析器实例化。

A. 任务定义

我们将 SignNav 建模为部分可观测马尔可夫决策过程 (POMDP)。目标是让无身体代理在连续 3D 环境中, 通过利用从标识牌观察到的与目标相关的方向线索, 导航至语义目标 g 。

观测空间: 在每个时间步 t , 智能体接收一个以自我为中心的 RGB-D 观测 $o_t = \{o_t^r, o_t^d\}$ 。上游符号解析器提取方向线索及其图像位置, 如下所示:

$$\psi(o_t, g) \rightarrow (c_t, b_t) \quad (1)$$

其中 c_t 表示解析后的目标相关提示, $b_t = [x_{min}, y_{min}, x_{max}, y_{max}]$ 表示其边界框。提示 c_t 可根据解析器接口表示为视觉裁剪或结构化方向类别。

动作空间: 智能体使用离散的 $a_t \in \mathcal{A}$ 动作空间, 包含四个低级命令: 前进 (0.25 米)、左转 (15°)、右转 (15°) 和停止。该设置遵循常见的连续环境导航协议 [15]。

策略学习: 由于标志提示稀疏, 且可能在对应的连接点变得可操作之前就被观察到, 因此智能体必须利用历史上下文。我们将历史定义为:

$$\mathcal{H}_t \triangleq \{(o_0, c_0, b_0, a_0), \dots, (o_{t-1}, c_{t-1}, b_{t-1}, a_{t-1})\} \quad (2)$$

目标是学习一个由 Θ 参数化的策略 π , 该策略根据以下规则预测下一个导航动作:

$$a_t \sim \pi(a_t | \mathcal{H}_t, o_t, c_t, b_t; \Theta) \quad (3)$$

当代理在目标附近发出 Stop 指令或达到最大步数限制时, 该回合结束。

B. LSI 数据集创建

现有具身导航数据集 [32], [33] 主要集中在家庭规模环境中, 未能充分捕捉公共室内空间的空间尺度与拓扑结构。为支持 SignNav, 我们利用从开源仓库 [32], [34], [35], 收集的高保真 3D 扫描数据, 构建了大规模室内 (LSI) 数据集, 涵盖医院、办公室及其他大型室内环境。这些场景包含长走廊、开放大厅及复杂的拓扑结构, 非常适合评估长时程的基于标志线索的导航任务。

标识注入: 我们开发了一个基于 Blender 的自动化管道, 以过程化方式将标识注入到 3D 网格中。候选标识位置被选择在路口和拐角等决策点周围, 并根据局部路线几何形状分配方向提示。为了减少过拟合到单一视觉模板, 数据集包含多样化的标识外观, 包括箭头形状、颜色、布局 and 背景风格的变体。图 2 展示了医院场景中的渲染效果。在本工作中, 我们专注于方向提示, 并将更丰富的标识理解(如 OCR 和复合指令解析)留给上游感知模块。

Cue Representation: 对于每个可见的目标相关标志, 数据集提供了两种互补形式的方向提示: (1) 视觉提示, 由方向符号的裁剪图像区域表示; (2) 结构化提示, 由解析出的方向类别表示, 例如 *left* 或 *forward*。两种提示形式均与相同的图像空间边界框 b_t 配对, 使模型能够在统一的导航协议下研究视觉提示定位或基于解析的结构化提示定位。

场景生成: 针对每个场景, 我们使用泊松盘采样 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ 构建可导航的连接图 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ 。从图节点 $\mathcal{V}$ 中采样起点 - 终点对, 并使用 Dijkstra 算法规划最短路径, 随后进行基于样条的平滑处理。随后, 一个代理 (oracle agent) 执行平滑后的轨迹以生成训练和评估场景。为了自动标注每个场景中的方向标志提示, 我们通过利用平滑轨迹的局部曲率和代理相对于注入标志的朝向来确定代理提示方向。在每个时间步 t , 我们记录 RGB-D 观测 o_t 、与目标相关的方向提示 c_t 、其边界框 b_t 以及代理动作 a_t^* :

$$\tau = \{(o_t, c_t, b_t, a_t^*)\}_{t=0}^T \quad (4)$$

其中 T 为 episode horizon。在仿真中, c_t 和 b_t 来自过程注释, 为研究下游方向线索引导导航提供了受控接口。

IV. SPATIAL-TEMPORAL AWARE TRANSFORMER

Fig. 3 illustrates the architecture of the proposed START model for directional signage-cue guided navigation. At each time step t , START receives the egocentric RGB-D observation o_t , a target-relevant directional cue c_t , and its bounding box b_t from the upstream parser. The cue can be represented either as a visual crop or as a structured direction category. START first encodes the observation and cue, then uses a *Spatial-Aware Transformer* to ground the cue in the current egocentric scene. To handle sparse or delayed signage observations, a *Temporal-Aware Transformer* further aggregates historical states and actions. The resulting state representation is finally passed to an action head to predict the next discrete navigation action.

A. Input Embeddings

Observation Embedding: Given the RGB-D observation $o_t = \{o_t^r, o_t^d\}$, we encode the RGB image and depth image using two lightweight CNN encoders with the same architecture but different input channels. The resulting features, v_t^r and v_t^d , are fused and normalized to obtain the observation embedding:

$$v_t^o = \text{LayerNorm}(v_t^r + v_t^d) \quad (5)$$

This embedding provides both appearance and geometric information for subsequent cue grounding.

Cue Embedding: The target-relevant directional cue c_t can be represented in two forms: a visual cue crop or a structured direction category. For the visual-cue interface, we encode the cropped cue image using a ResNet-50 [37] pre-trained on ImageNet. For the structured-cue interface, we encode the parsed direction category with a learnable category embedding. In both cases, the cue is paired with its image-space bounding box b_t . We normalize the box coordinates as $f_t^b = [\frac{x_{min}}{W}, \frac{y_{min}}{H}, \frac{x_{max}}{W}, \frac{y_{max}}{H}, \frac{w}{W}, \frac{h}{H}]$, where W and H are the image width and height. The final cue embedding is obtained by projecting and fusing the cue feature f_t^c and the box feature f_t^b as:

$$v_t^c = \text{LayerNorm}(f_t^c W^c) + \text{LayerNorm}(f_t^b W^b) \quad (6)$$

where W^c and W^b are learnable projection matrices. This design allows START to use either visual cue grounding or parser-based structured cue grounding under the same spatial-temporal policy architecture.

B. Spatial-Aware Transformer

Fig. 4 illustrates the navigation action decisions made by the agent leveraging directional signage-cue under different circumstances. The same directional cue may lead to different actions depending on the agent's pose, cue location, and visible scene structure. Conversely, different cues observed at the same location may require different navigation decisions. Therefore, START must ground the target-relevant cue in the current egocentric observation rather than directly treating the cue as an executable action command.

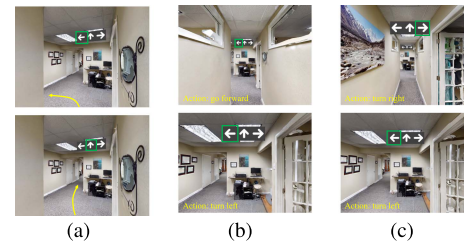


Fig. 4. Different action decisions under different circumstances. (a) different directional cues at the same location; (b) the same directional cues at different locations; (c) different directional cues at different locations.



Fig. 5. One period in an episode that there is no visible directional cue of signage. The agent still needs to make correct decisions by incorporating historical states.

Given the observation embedding v_t^o and cue embedding v_t^c , we use a *Spatial-Aware Transformer* to model their interaction inspired by the Vision Transformer (ViT) [38]. We first reshape v_t^o into a sequence of flattened visual patches $v_t^{patch} \in \mathbb{R}^{N \times (P^2 \cdot C_o)}$, where P is the patch size and N is the number of patches. These patches are projected into a D -dimensional latent space. Instead of initializing the [CLS] token independently, we initialize it with the projected cue embedding v_t^c , so that the cue serves as the query for retrieving relevant visual regions. The self-attention blocks in *Spatial-Aware Transformer* is computed as follows:

$$Q = X_{l-1} W_{l,k}^Q, K = X_{l-1} W_{l,k}^K, V = X_{l-1} W_{l,k}^V \quad (7)$$

$$H_{l,k} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right) V \quad (8)$$

where W^Q , W^K , and W^V are learnable projections, and d_h is the attention head dimension. The resulting patch features form the spatial embedding f_t^{spa} , while the updated cue token summarizes the cue-conditioned scene context for subsequent temporal reasoning. This module is shared by both visual-cue and structured-cue interfaces.

C. Temporal-Aware Transformer

In LSI environments, directional signage cues are not continuously visible. A cue may disappear because of the limited field of view, or it may be observed before the corresponding junction becomes actionable (as shown in Fig. 5). Therefore, the agent must retain previously observed cue-grounded information and combine it with the current observation. To this end, START uses a *Temporal-Aware Transformer* to model dependencies across recent navigation states.

IV. 时空感知 Transformer

图 3 展示了为方向标志线索引导导航而提出的 START 模型的架构。在每个时间步 t , START 接收来自上游解析器的自中心 RGB-D 观测 o_t 、与目标相关的方向线索 c_t 及其边界框 b_t 。该线索可以表示为视觉裁剪或结构化方向类别。START 首先对观测和线索进行编码, 然后利用空间感知 Transformer 将线索定位到当前的自中心场景中。为了处理稀疏或延迟的标志观测, 时间感知 Transformer 进一步聚合历史状态和行动。最终得到的状态表示被传递给行动头, 以预测下一个离散的导航行动。

A. 输入嵌入

观测嵌入: 给定 RGB-D 观测 $o_t = \{o_t^r, o_t^d\}$, 我们使用两个具有相同架构但输入通道数不同的轻量级 CNN 编码器对 RGB 图像和深度图像进行编码。所得特征 v_t^r 和 v_t^d 被融合并归一化, 以获得观测嵌入:

$$v_t^o = \text{LayerNorm}(v_t^r + v_t^d) \quad (5)$$

该嵌入为后续提示定位同时提供外观和几何信息。

Cue Embedding: 与目标相关的方向提示 c_t 可以以两种形式表示: 视觉提示裁剪图或结构化方向类别。对于视觉提示接口, 我们使用在 ImageNet 上预训练的 ResNet-50 [37] 对裁剪的提示图像进行编码。对于结构化提示接口, 我们使用可学习的类别嵌入对解析出的方向类别进行编码。在这两种情况下, 提示均与其图像空间边界框 b_t 配对。我们将边界框坐标归一化为 $f_t^b = [\frac{x_{min}}{W}, \frac{y_{min}}{H}, \frac{x_{max}}{W}, \frac{y_{max}}{H}, \frac{w}{W}, \frac{h}{H}]$, 其中 W 和 H 分别为图像宽度和高度。最终的提示嵌入通过将提示特征 f_t^c 和边界框特征 f_t^b 进行投影和融合获得, 公式如下:

$$v_t^c = \text{LayerNorm}(f_t^c W^c) + \text{LayerNorm}(f_t^b W^b) \quad (6)$$

其中 W^c 和 W^b 是可学习的投影矩阵。该设计使得 START 能够在相同的时空策略架构下, 使用视觉线索定位或基于解析器的结构化线索定位。

B. 空间感知 Transformer

图 4 展示了代理在不同情境下利用方向性标志线索所做出的导航动作决策。相同的方向性线索可能导致不同的动作, 这取决于代理的姿态、线索位置以及可见的场景结构。反之, 在同一位置观察到的不同线索可能需要不同的导航决策。因此, START 必须将目标相关的线索锚定在当前以自我为中心的观测中, 而不是直接将线索视为可执行的动作命令。

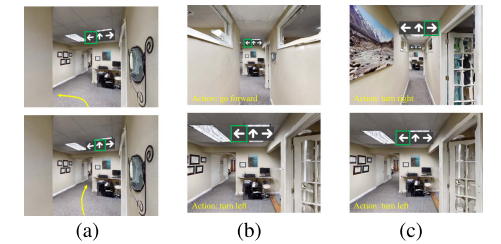


图 4. 不同情境下的不同行动决策。(a) 同一位置的不同方向线索; (b) 不同位置相同的方向线索; (c) 不同位置的不同方向线索。



图 5. 一个没有可见方向标志的周期。代理仍然需要通过结合历史状态来做出正确决策。

给定观察嵌入 v_t^o 和提示嵌入 v_t^c , 我们受视觉 Transformer (ViT) [38] 的启发, 使用空间感知 Transformer 来建模它们的交互。首先, 我们将 v_t^o 重塑为扁平化视觉片段的序列 $v_t^{patch} \in \mathbb{R}^{N \times (P^2 \cdot C_o)}$, 其中 P 是片大小, N 是片段数量。这些片段被投影到一个 D 维的潜在空间中。与独立初始化 [CLS] 标记不同, 我们使用投影后的提示嵌入 v_t^c 对其进行初始化, 从而使提示作为查询以检索相关的视觉区域。空间感知 Transformer 中的自注意力块计算如下:

$$Q = X_{l-1} W_{l,k}^Q, K = X_{l-1} W_{l,k}^K, V = X_{l-1} W_{l,k}^V \quad (7)$$

$$H_{l,k} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right) V \quad (8)$$

其中 W^Q 、 W^K 和 W^V 是可学习的投影, d_h 是注意力头维度。生成的 patch 特征构成空间嵌入 f_t^{spa} , 而更新后的 cue token 则总结用于后续时序推理的 cue 条件场景上下文。该模块同时被视觉 cue 接口和结构化 cue 接口共享。

C. 时序感知 Transformer

在 LSI 环境中, 方向性标志线索并非持续可见。线索可能因视野有限而消失, 也可能在对应的路口变为可操作之前就被观察到 (如图 5 所示)。因此, 代理必须保留先前观察到的基于线索的信息, 并将其与当前观察相结合。为此, START 使用时间感知 Transformer 来建模最近导航状态之间的依赖关系。



Fig. 6. Trajectories visualization of our method compared to baselines. The second row shows the first-person-view of agent as it navigates along the ground-truth trajectory.

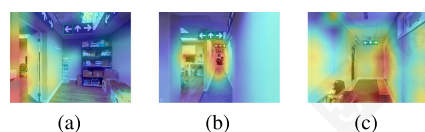


Fig. 7. Attention map visualization both with the spatial-aware and temporal-aware modules in our START model. The darker the color, the greater the attention weight.

Let $\mathcal{H}_k = \{(f_{t-k}^{spa}, a_{t-k}), \dots, (f_{t-1}^{spa}, a_{t-1})\}$ denote a fixed-length history buffer, where f_i^{spa} is the spatial embedding produced by the Spatial-Aware Transformer and a_i is the executed action. For each historical step i , we aggregate f_i^{spa} by average pooling and fuse it with the action embedding:

$$s_i = \text{LayerNorm}(e_i^{spa} \mathbf{W}^{spa}) + \text{LayerNorm}(e_i^a \mathbf{W}^a) \quad (9)$$

where e_i^{spa} denotes the pooled spatial feature and e_i^a denotes the embedded action.

The sequence of historical states $\{s_{t-k}, \dots, s_{t-1}\}$, together with the current cue-conditioned state s_t , are then processed by *Temporal-Aware Transformer*. The k -th attention head at the l -th layer performs self-attention over $\mathbf{X}_{l-1}$ following (7) and (8). The outputs from all the attention heads will be concatenated and projected onto the same dimension as the input as:

$$\mathbf{H}_l = [\mathbf{H}_{l,1}; \dots; \mathbf{H}_{l,k}] \mathbf{W}_l^o \quad (10)$$

where k is the total number of attention heads, $[\cdot]$ denotes concatenation. Finally, the output of layer l is formulated by:

$$\mathbf{H}_l' = \text{LayerNorm}(\mathbf{H}_l + \mathbf{X}_{l-1}) \quad (11)$$

$$\mathbf{X}_l' = \text{GeLU}(\mathbf{H}_l' \mathbf{W}_l^{F1}) \mathbf{W}_l^{F2} \quad (12)$$

$$\mathbf{X}_l = \text{LayerNorm}(\mathbf{H}_l' + \mathbf{X}_l') \quad (13)$$

where GeLU is the Gaussian Error Linear Unit activation function. The final state representation f_t^s encodes both the current cue-grounded observation and recent navigation history, enabling the policy to act when signage is temporarily unavailable or when a previously observed cue must be executed later.

D. Training Objective

We train START as a closed-loop action prediction policy for directional signage-cue guided navigation. Given the state representation f_t^s produced by the spatial-temporal modules, an MLP action head predicts the probability distribution over the discrete action space:

$$p_t^a = \text{Softmax}(f_t^s \mathbf{W}^s) \quad (14)$$

where $\mathbf{W}^s$ denotes learnable parameters. The policy is first optimized by imitation learning from oracle trajectories. We minimize the inflection-weighted cross-entropy loss:

$$\mathcal{L}_{loss} = - \sum_t w_t a_t^* \log(p_t^a) \quad (15)$$

where a_t^* is the oracle action and w_t is the inflection weight, which emphasizes timesteps where the expert action changes.

To reduce covariate shift during closed-loop execution, we further fine-tune the policy using Dataset Aggregation (DAgger) [39]. At each iteration, the current policy interacts with the environment, and the resulting states are relabeled by the oracle and added to the training set. This two-stage training strategy improves robustness when the agent deviates from the reference trajectory.

V. EXPERIMENTS

A. Experimental Setup

Dataset: We train and evaluate all methods in Habitat on the LSI-Dataset. The dataset contains 20 LSI environments split into train, val-seen, and val-unseen sets with 16, 16, and 4 environments, respectively. Environments in val-seen split are the same as training, while the val-unseen split contains held-out environments. We collect 8000, 3200, and 800 episodes for the three splits, with an average trajectory length of approximately 10 m. The dataset includes diverse sign appearances and provides each target-relevant directional cue in two forms: a visual crop and a structured direction category, both paired with the cue bounding box.

Baselines: We compare START with three representative baselines in the form of visual crop as cue representation. The **Rule-based** baseline [25] follows a modular signage-guided navigation pipeline and is provided with ground-truth directional cues and local metric-topological information for subgoal selection. The **VLM-based** baseline uses Qwen3 [40] to reason navigation actions from the current observation and parsed sign cues; the exact prompt, action set, and decoding rules are provided in the supplementary material. The **ViNT** baseline [41] is adapted to **SignNav** by replacing its goal encoder with a signage-cue encoder and retrained under the same dataset split. These baselines cover rule-based planning, vision-language reasoning, and learning-based visual navigation.

Metrics: We report Success Rate (SR), normalized Dynamic Time Warping (NDTW), and Success weighted by normalized Dynamic Time Warping (SDTW) [1], [42]. SR measures whether the agent stops within 1.0 m of the target, while NDTW and SDTW evaluate trajectory similarity and successful path following. In addition, we report Correct Sign-Following Length



图 6. 展示了我们方法与基线方法的轨迹可视化。第二行展示了代理沿真实轨迹导航时的一人称视角。

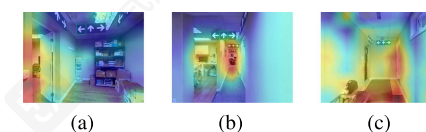


图 7. 展示了 START 模型中空间感知模块和时间感知模块的注意力图可视化。颜色越深, 表示注意力权重越大。

设 $\mathcal{H}_k = \{(f_{t-k}^{spa}, a_{t-k}), \dots, (f_{t-1}^{spa}, a_{t-1})\}$ 为固定长度的历史缓冲区, 其中 f_i^{spa} 是由空间感知 Transformer 生成的空间嵌入, a_i 为执行的动作。对于每个历史步骤 i , 我们通过平均池化聚合 f_i^{spa} , 并将其与动作嵌入融合:

$$s_i = \text{LayerNorm}(e_i^{spa} \mathbf{W}^{spa}) + \text{LayerNorm}(e_i^a \mathbf{W}^a) \quad (9)$$

其中 e_i^{spa} 表示池化后的空间特征, e_i^a 表示嵌入的动作。

历史状态序列 $\{s_{t-k}, \dots, s_{t-1}\}$ 与当前提示条件状态 s_t 随后由时间感知 Transformer 进行处理。第 l 层的第 k 个注意力头会对 (7) 和 (8) 之后的 $\mathbf{X}_{l-1}$ 个数据进行自注意力操作。所有注意力头的输出将被拼接并投影到与输入相同的维度, 如下所示:

$$\mathbf{H}_l = [\mathbf{H}_{l,1}; \dots; \mathbf{H}_{l,k}] \mathbf{W}_l^o \quad (10)$$

其中 k 是注意力头的总数, $[\cdot]$ 表示拼接。最后, 层 l 的输出由下式给出:

$$\mathbf{H}_l' = \text{LayerNorm}(\mathbf{H}_l + \mathbf{X}_{l-1}) \quad (11)$$

$$\mathbf{X}_l' = \text{GeLU}(\mathbf{H}_l' \mathbf{W}_l^{F1}) \mathbf{W}_l^{F2} \quad (12)$$

$$\mathbf{X}_l = \text{LayerNorm}(\mathbf{H}_l' + \mathbf{X}_l') \quad (13)$$

其中 GeLU 是高斯误差线性单元激活函数。最终状态表示 f_t^s 编码了当前的线索 - 地面观察和最近的导航历史, 使策略能够在标志牌暂时不可用时或先前观察到的线索需要稍后执行时采取行动。

D. 训练目标

我们训练 START 为一种闭环动作预测策略, 用于方向标志线索引导的导航。给定由时空模块生成的状态表示 f_t^s , 一个 MLP 动作头预测离散动作空间上的概率分布:

$$p_t^a = \text{Softmax}(f_t^s \mathbf{W}^s) \quad (14)$$

其中 $\mathbf{W}^s$ 表示可学习参数。策略首先通过从专家轨迹进行模仿学习进行优化。我们最小化拐点加权交叉熵损失:

$$\mathcal{L}_{loss} = - \sum_t w_t a_t^* \log(p_t^a) \quad (15)$$

其中 a_t^* 是预言机动作, w_t 是曲率权重, 用于强调专家动作发生变化的时间步。

为减少闭环执行期间的协变量偏移, 我们进一步使用数据集聚合 (DAgger) [39] 微调策略。在每次迭代中, 当前策略与环境交互, 生成的状态由预言机重新标记并加入训练集。这种两阶段训练策略提高了当智能体偏离参考轨迹时的鲁棒性。

V. 实验

A. 实验设置

数据集: 我们在 Habitat 上所有方法均在 LSI-Dataset 上进行训练和评估。该数据集包含 20 个 LSI 环境, 分为训练集、验证已知集和验证未知集, 分别包含 16、16 和 4 个环境。验证已知集中的环境与训练集相同, 而验证未知集包含保留测试的环境。我们为三个划分分别收集了 8000、3200 和 800 个 episode, 平均轨迹长度约为 10 米。该数据集包含多样化的标志外观, 并为每个与目标相关的方向提示提供两种形式: 视觉裁剪和结构化方向类别, 两者均与提示边界框配对。

基线: 我们将 START 与三种以视觉裁剪作为线索表示的代表性基线进行对比。基于规则的基线 [25] 遵循模块化标志引导导航流程, 并提供真实的方向线索和局部度量拓扑信息用于子目标选择。基于 VLM 的基线使用 Qwen3 [40] 根据当前观测和解析的标志线索推理导航动作; 具体提示、动作集和解码规则见补充材料。ViNT 基线 [41] 通过将其目标编码器替换为标志线索编码器并基于相同数据集划分重新训练, 适配到 SignNav。这些基线涵盖了基于规则的规划、视觉 - 语言推理以及基于学习的视觉导航。

指标: 我们报告成功率 (SR)、归一化动态时间规整 (NDTW) 以及按归一化动态时间规整加权的成功率 (SDTW) [1], [42]。SR 衡量代理是否在距离目标 1.0 米范围内停止, 而 NDTW 和 SDTW 评估轨迹相似性和成功路径跟随。此外, 我们报告正确的符号跟随长度。

TABLE I
NAVIGATION PERFORMANCE COMPARISON WITH BASELINES

Methods	Val-Seen				Val-Unseen			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
VLM-based	0.35	0.39	0.27	0.50	0.42	0.47	0.34	0.53
Rule-based [25]	0.38	0.57	0.31	0.46	0.42	0.58	0.35	0.57
ViNT [41]	0.59	0.52	0.48	0.60	0.46	0.42	0.38	0.53
START (w/o DAgger)	0.65	0.59	0.52	0.67	0.49	0.47	0.38	0.57
START (w/ DAgger)	0.83	0.72	0.68	0.77	0.57	0.53	0.44	0.62

TABLE II
EFFECT OF CUE REPRESENTATION AND BOUNDING-BOX ABLATION

Cue Representation	BBox	Val-Unseen			
		SR	NDTW	SDTW	CSFL
Visual (w/o DAgger)	w/ bbox w/o bbox	0.49 0.43	0.47 0.42	0.38 0.34	0.57 0.52
Category (w/o DAgger)	w/ bbox w/o bbox	0.61 0.33	0.58 0.36	0.49 0.25	0.65 0.51
No semantic cue	w/ bbox	0.38	0.34	0.29	0.48

(CSFL), which measures the number of ground-truth signs correctly followed before the agent gets off the reference path. CSFL complements trajectory-level metrics by directly evaluating how long the agent remains consistent with the intended signage sequence.

Implementation details: For START, RGB and depth observations are encoded by lightweight CNN encoders, and the signage cue is encoded from the visual crop using ResNet-50 [37]. In the experiment of cue representation, the signage cue in the form of structured direction category is encoded using a learnable embedding. The Spatial-Aware Transformer adopts a ViT-B/32-style architecture [38], and the Temporal-Aware Transformer follows a BERT-like Transformer design [43]. Both modules use hidden size 768, 6 layers, and 6 attention heads. We first train the policy with teacher forcing and then fine-tune it using DAgger [39]. All simulation evaluations are conducted on one NVIDIA RTX 4090 GPU.

B. Main Results

Navigation performance: Table I compares START with representative baselines on the LSI-Dataset. START achieves the best overall performance on both val-seen and val-unseen splits. Compared with ViNT, START with DAgger improves SR from 0.59 to 0.83 on val-seen and from 0.46 to 0.57 on val-unseen, while also improving NDTW, SDTW, and CSFL. The rule-based baseline uses hand-crafted decisions and local topological information, but its performance remains limited by fixed heuristics and scene-dependent route structures. The VLM-based baseline can recognize directional cues in many cases, but its closed-loop low-level action prediction is unstable. ViNT provides a stronger learning-based baseline, yet lacks explicit cue-observation grounding and temporal memory. The improvement of START, especially in CSFL, indicates that the proposed spatial-temporal policy follows the intended signage sequence more consistently before deviating from the reference route. DAgger further improves closed-loop robustness by reducing the effect of covariate shift during execution.

Effect of cue representation: Table II studies different cue interfaces and the contribution of bounding-box information on the val-unseen split. The structured direction category with

TABLE III
QUANTITATIVE RESULTS OF REAL-WORLD DEPLOYMENT

Metrics	Lifts		Restroom		Pantry		Avg.
	1	2	3	4	5	6	
Success	1	1	1	0	1	1	5 / 6
Moved Distance (m)	26.1	25.8	15.3	23.4	15.7	15.3	20.3
Total Time (s)	212	201	133	173	121	104	157

TABLE IV
RUNTIME OF EACH MODULE IN THE DEPLOYMENT SYSTEM

Modules	Avg. Time (ms)	Avg. Frequency (Hz)
Transmission (RGB)	~280	3.57
Qwen3	~1800	0.55
DepthAnything-V2	~110	9.10
START Inference	~25	40.00
Transmission (Action)	~5	200.00
Total	2220	0.45

bounding box achieves the best performance, improving SR from 0.49 to 0.61 over the visual-crop cue with bounding box. This indicates that, when an upstream parser is available, representing the signage cue as a direction category can reduce dependence on low-level arrow appearance and improve generalization. Nevertheless, we use the visual crop as the default cue interface in the main experiments and subsequent ablation studies, since it preserves the original visual-cue grounding setting and directly evaluates whether START can learn from iconic signage observations. Removing the bounding box degrades both visual and category-based variants, showing that cue location is important for grounding the directional cue in the egocentric scene. The no-cue variant performs worse than the full cue-conditioned models, confirming that START benefits from both cue semantic and spatial cue localization rather than relying only on scene geometry.

Real-world deployment: We deploy START on a DeepRobotics Lite2 robot equipped with an Intel RealSense D435i camera in an office environment. At each step, the robot captures and transmits the RGB image to a remote desktop through wireless LAN. Qwen3-8B [40] is used as the upstream sign parser to extract the target-relevant directional cue and its crop location from the RGB observation, while DepthAnything-V2 [44] is used to generate a refined depth image for more stable geometric perception. START then takes the RGB image, generated depth image, parsed cue, and cue location as input, and predicts a discrete navigation action that is transmitted back to the robot for execution.

As shown in Table III, START succeeds in 5 out of 6 trials across three target types: lifts, restroom, and pantry. The robot travels 20.3 m on average and spends 157 s per trial. These quantitative trials complement the qualitative demonstration and show that the proposed downstream policy can use target-relevant signage cues in physical environments. Table IV reports the runtime of each deployment module. START inference runs at approximately 40 Hz, while the full system operates at about 0.45 Hz. The main latency comes from WLAN RGB transmission and Qwen3-8B signage parsing rather than from START itself, suggesting that faster onboard sign parsing and communication will be important for deployment in more dynamic public environments.

表 I 与基线导航性能对比

Methods	Val-Seen				Val-Unseen			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
VLM-based	0.35	0.39	0.27	0.50	0.42	0.47	0.34	0.53
Rule-based [25]	0.38	0.57	0.31	0.46	0.42	0.58	0.35	0.57
ViNT [41]	0.59	0.52	0.48	0.60	0.46	0.42	0.38	0.53
START (w/o DAgger)	0.65	0.59	0.52	0.67	0.49	0.47	0.38	0.57
START (w/ DAgger)	0.83	0.72	0.68	0.77	0.57	0.53	0.44	0.62

表 II 提示表示与边界框消融的影响

Cue Representation	BBox	Val-Unseen			
		SR	NDTW	SDTW	CSFL
Visual (w/o DAgger)	w/ bbox w/o bbox	0.49 0.43	0.47 0.42	0.38 0.34	0.57 0.52
Category (w/o DAgger)	w/ bbox w/o bbox	0.61 0.33	0.58 0.36	0.49 0.25	0.65 0.51
No semantic cue	w/ bbox	0.38	0.34	0.29	0.48

(CSFL), 用于衡量在智能体离开参考路径之前正确遵循的地面真值标志数量。CSFL 通过直接评估智能体与预期标志序列保持一致的时间长度, 补充了轨迹级指标。

实现细节: 对于 START、RGB 和深度观测, 由轻量级 CNN 编码器进行编码, 而标志线索则通过 ResNet-50 [37] 对视觉裁剪进行编码。在线索表示实验中, 结构化方向类别形式的标志线索使用可学习嵌入进行编码。空间感知 Transformer 采用 ViT-B/32 风格的架构 [38], 时间感知 Transformer 遵循类似 BERT 的 Transformer 设计 [43]。这两个模块均使用隐藏层大小 768、6 层和 6 个注意力头。我们首先使用教师强制训练策略, 然后使用 DAgger [39] 进行微调。所有仿真评估均在单张 NVIDIA RTX 4090 GPU 上进行。

B. 主要结果

导航性能: 表 I 在 LSI 数据集上将 START 与代表性基线进行了比较。START 在 val-seen 和 val-unseen 两个划分上均取得了最佳整体性能。与 ViNT 相比, 带有 DAgger 的 START 在 val-seen 上使 SR 从 0.59 提升至 0.83, 在 val-unseen 上使 SR 从 0.46 提升至 0.57, 同时 NDTW、SDTW 和 CSFL 指标也得到改善。基于规则的基线使用手工构造的决策和局部拓扑信息, 但其性能仍受限于固定启发式规则和场景相关的路线结构。基于 VLM 的基线虽然在许多情况下能够识别方向线索, 但其闭环低层动作预测不稳定。ViNT 提供了更强的基于学习的基线, 但缺乏显式的线索 - 观测对齐和时序记忆。START 的提升, 尤其是在 CSFL 上的表现, 表明所提出的时空策略在偏离参考路线之前, 能更一致地遵循预期的标志序列。DAgger 通过在执行过程中减少协变量偏移的影响, 进一步提升了闭环鲁棒性。

提示表征的影响: 表 II 研究了不同的提示接口以及边界框信息在 val-unseen 划分中的贡献。带有结构化方向类别的

表 III 真实世界部署的定量结果

Metrics	Lifts		Restroom		Pantry		Avg.
	1	2	3	4	5	6	
Success	1	1	1	0	1	1	5 / 6
Moved Distance (m)	26.1	25.8	15.3	23.4	15.7	15.3	20.3
Total Time (s)	212	201	133	173	121	104	157

表 IV 部署系统中各模块的运行时间

Modules	Avg. Time (ms)	Avg. Frequency (Hz)
Transmission (RGB)	~280	3.57
Qwen3	~1800	0.55
DepthAnything-V2	~110	9.10
START Inference	~25	40.00
Transmission (Action)	~5	200.00
Total	2220	0.45

边界框实现了最佳性能, 相比仅使用视觉裁剪提示, 将显著性重定向 (SR) 从 0.49 提升至 0.61。这表明, 当存在上游解析器时, 将标志提示表示为方向类别可以减少对低级箭头外观的依赖, 并提升泛化能力。尽管如此, 我们在主要实验和后续消融研究中仍采用视觉裁剪作为默认提示接口, 因为它保留了原始视觉提示的锚定设置, 并直接评估 START 是否能够从具象化标志观察中学习。移除边界框会导致基于视觉和基于类别的变体性能均下降, 表明提示位置对于在自中心场景中锚定方向提示至关重要。无提示变体的表现不如全提示条件模型, 证实 START 受益于提示的语义和空间定位, 而不仅仅依赖场景几何结构。

真实场景部署: 我们在办公室环境中, 将 START 部署在一台配备 Intel RealSense D435i 摄像头的 DeepRobotics Lite2 机器人上。在每一步中, 机器人捕获并经由无线局域网将 RGB 图像传输至远程桌面。Qwen3-8B [40] 被用作上游符号解析器, 从 RGB 观测中提取与目标相关的方向线索及其裁剪位置, 同时 DepthAnything-V2 [44] 用于生成精细的深度图, 以实现更稳定的几何感知。随后, START 将 RGB 图像、生成的深度图、解析出的线索及其位置作为输入, 预测一个离散的导航动作, 并将其传回机器人执行。

如表 III 所示, START 在三种目标类型 (电梯、卫生间和储藏室) 的六次三轮测试中成功 5 次。机器人平均行进 20.3 米, 每次测试耗时 157 秒。这些定量测试补充了定性演示, 表明所提出的下游策略能够在物理环境中利用与目标相关的标识线索。表 IV 报告了每个部署模块的运行时间。START 推理频率约为 40 Hz, 而完整系统的运行频率约为 0.45 Hz。主要延迟来自 WLAN RGB 传输和 Qwen3-8B 标识解析, 而非 START 本身, 这表明更快的车载标识解析和通信对于在更动态的公共环境中部署至关重要。

TABLE V
ABLATION STUDY OF SPATIAL- AND TEMPORAL-AWARE MODULES

Spatial-Aware	Temporal-Aware	Val-Unseen			
		SR	NDTW	SDTW	CSFL
×	✓	0.35	0.39	0.28	0.49
✓	×	0.39	0.41	0.32	0.48
✓	✓	0.49	0.47	0.38	0.57

TABLE VI
ABLATION STUDY UNDER DIFFERENT SIGNAGE-CUE DROPOUT RATIOS

Dropout Ratio	w/ Temporal-Aware				w/o Temporal-Aware			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
0.0	0.49	0.47	0.38	0.57	0.39	0.41	0.32	0.48
0.2	0.47	0.45	0.37	0.55	0.40	0.42	0.33	0.50
0.4	0.47	0.44	0.37	0.54	0.39	0.40	0.32	0.48
0.6	0.46	0.44	0.36	0.53	0.34	0.39	0.28	0.46
0.8	0.42	0.42	0.33	0.51	0.27	0.33	0.22	0.40

C. Ablation Studies

How important are the spatial- and temporal-aware modules? Table V reports the module ablation results on the val-unseen split. Removing either module degrades performance, while the full START model achieves the best results across all metrics. The spatial-only variant performs slightly better than the temporal-only variant in SR, NDTW, and SDTW, indicating that grounding the directional cue in the current egocentric observation is essential for action prediction. However, the full model further improves SR from 0.39 to 0.49 and CSFL from 0.48 to 0.57, showing that temporal reasoning helps the agent follow the intended signage sequence more consistently. These results support the complementary roles of spatial grounding and temporal memory in directional signage-cue guided navigation.

How does bounding-box position affect spatial grounding? We analyze the effect of bounding-box position using the ablation in Table II. Removing the bounding box consistently degrades performance for both cue interfaces. For the visual-crop cue, SR decreases from 0.49 to 0.43 and CSFL decreases from 0.57 to 0.52. For the structured category cue, the degradation is more pronounced, with SR dropping from 0.61 to 0.33 and SDTW from 0.49 to 0.25. These results indicate that the cue semantics alone are insufficient for robust navigation: the agent also needs the image-space cue location to ground the directional instruction in the current egocentric scene. The bounding box therefore provides important spatial information rather than merely serving as an auxiliary input.

How robust is START to sparse signage cues? Table VI evaluates the effect of temporal memory under different signage-cue dropout ratios. Specifically, we simulate different levels of signage sparsity by randomly dropping out the ground-truth signage observations along each episode with a certain probability. As the dropout ratio increases, the performance of both variants decreases, indicating that visible signage cues are important for accurate navigation. However, the model with the temporal-aware module consistently outperforms the variant without it across all dropout settings. The advantage becomes more evident under severe sparsity: at a dropout ratio of 0.8, the temporal-aware model improves SR from 0.27 to 0.42 and CSFL from 0.40 to 0.51. These results show that temporal memory helps START retain previously observed directional cues and maintain

TABLE VII
ABLATION STUDY OF DELAYED-TURN SETTINGS

Settings	Val-Unseen			
	SR	NDTW	SDTW	CSFL
Original	0.49	0.47	0.38	0.57
Delayed Turn (0.5 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.0 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.5 m)	0.50	0.47	0.39	0.57
Delayed Turn (2.0 m)	0.46	0.42	0.35	0.53

TABLE VIII
ABLATION STUDY OF DIFFERENT INPUT MODALITIES

RGB	Depth	Val-Unseen			
		SR	NDTW	SDTW	CSFL
✓	×	0.41	0.42	0.31	0.54
×	✓	0.56	0.52	0.45	0.62
✓	✓	0.49	0.47	0.38	0.57

the intended signage-following behavior when current cues are missing.

Can START handle delayed turning after observing signage? Table VII evaluates whether START can execute a turn after the signage cue has been observed earlier than the actual junction. In this setting, the current signage cue is removed within a specified distance before the junction, forcing the agent to rely on its previous cue observation and current scene geometry. START maintains nearly unchanged performance for delayed distances from 0.5 m to 1.5 m, with SR around 0.49 ~ 0.50 and CSFL remaining at 0.57. Performance decreases at 2.0 m, where the cue must be retained over a longer interval and larger trajectory uncertainty may accumulate. These results indicate that START can handle moderate delayed-turn cases through temporal memory and spatial grounding, while longer cue-action delays remain challenging.

What is the impact of different input modalities? Table VIII reports the effect of RGB and depth observations on START. The depth-only variant achieves the best performance, improving SR from 0.49 to 0.56 and CSFL from 0.57 to 0.62 compared with RGB-D input. This suggests that geometry provides strong cues for local traversability and route structure in the current LSI-Dataset. The RGB-only variant performs worse, indicating that appearance alone is less reliable for closed-loop control in these indoor scenes. Nevertheless, RGB remains necessary in the complete real-world system for upstream signage parsing and cue extraction. These results also highlight a limitation of the current benchmark: the policy may rely heavily on geometric structure, and future work should incorporate richer semantic supervision and more visually diverse environments where geometry alone is insufficient.

VI. CONCLUSION AND LIMITATIONS

This paper presented **SignNav**, a directional signage-cue guided visual navigation task for LSI environments. We constructed the **LSI-Dataset** with procedurally injected signage, diverse sign appearances, and two cue representations: visual cue crop and structured direction category. We also proposed **START**, a Spatial-Temporal Aware Transformer that grounds target-relevant directional cues in egocentric observations and uses temporal memory to handle sparse or delayed signage. Experiments on the LSI-Dataset, cue-representation ablations,

表 V
空间与时间感知模块消融研究

Spatial-Aware	Temporal-Aware	Val-Unseen			
		SR	NDTW	SDTW	CSFL
×	✓	0.35	0.39	0.28	0.49
✓	×	0.39	0.41	0.32	0.48
✓	✓	0.49	0.47	0.38	0.57

表 VI
不同标志提示丢弃比率下的消融研究

Dropout Ratio	w/ Temporal-Aware				w/o Temporal-Aware			
	SR	NDTW	SDTW	CSFL	SR	NDTW	SDTW	CSFL
0.0	0.49	0.47	0.38	0.57	0.39	0.41	0.32	0.48
0.2	0.47	0.45	0.37	0.55	0.40	0.42	0.33	0.50
0.4	0.47	0.44	0.37	0.54	0.39	0.40	0.32	0.48
0.6	0.46	0.44	0.36	0.53	0.34	0.39	0.28	0.46
0.8	0.42	0.42	0.33	0.51	0.27	0.33	0.22	0.40

C. 消融实验

空间感知与时序感知模块的重要性如何？表 V 报告了在 val-unseen 分割上的模块消融结果。移除任一模块均会导致性能下降，而完整的 START 模型在所有指标上均取得最佳结果。仅包含空间感知的变体在 SR、NDTW 和 SDTW 指标上略优于仅包含时序感知的变体，这表明将方向线索锚定在当前以自我为中心的观测中对于动作预测至关重要。然而，完整模型将 SR 从 0.39 提升至 0.49，将 CSFL 从 0.48 提升至 0.57，表明时序推理有助于代理更一致地遵循预期的标志序列。这些结果支持了空间锚定与时序记忆在方向标志线索引导导航中的互补作用。

边界框位置如何影响空间定位？我们利用表 II 中的消融实验分析了边界框位置的影响。移除边界框会一致地降低两种提示接口的性能。对于视觉裁剪提示，SR 从 0.49 降至 0.43，CSFL 从 0.57 降至 0.52。对于结构化类别提示，性能下降更为明显，SR 从 0.61 降至 0.33，SDTW 从 0.49 降至 0.25。这些结果表明，仅靠提示语义不足以实现鲁棒的导航：智能体还需要图像空间的提示位置，以便在当前以自我为中心的视图中对方向指令进行空间定位。因此，边界框提供了重要的空间信息，而不仅仅是作为辅助输入。

START 对稀疏路牌提示的鲁棒性如何？表 VI 评估了在不同路牌提示缺失比例下时序记忆的效果。具体而言，我们通过以一定概率随机丢弃每个回合中的真实路牌观测，来模拟不同级别的路牌稀疏性。随着缺失比例的增加，两个变体的性能均下降，表明可见路牌提示对于准确导航至关重要。然而，在所有缺失设置下，配备时序感知模块的模型始终优于未配备该模块的变体。这种优势在严重稀疏情况下尤为明显：在缺失比例为 0.8 时，时序感知模型将 SR 从 0.27 提升至 0.42，将 CSFL 从 0.40 提升至 0.51。这些结果表明，时序记忆有助于 START 保留先前观测到的方向提示并维持

表 VII 延迟转向设置的消融研究

Settings	Val-Unseen			
	SR	NDTW	SDTW	CSFL
Original	0.49	0.47	0.38	0.57
Delayed Turn (0.5 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.0 m)	0.49	0.47	0.38	0.57
Delayed Turn (1.5 m)	0.50	0.47	0.39	0.57
Delayed Turn (2.0 m)	0.46	0.42	0.35	0.53

表 VIII
不同输入模态的消融研究

RGB	Depth	Val-Unseen			
		SR	NDTW	SDTW	CSFL
✓	×	0.41	0.42	0.31	0.54
×	✓	0.56	0.52	0.45	0.62
✓	✓	0.49	0.47	0.38	0.57

当前线索缺失时的预期跟随标志行为。

START 能否在观察到路牌后执行延迟转弯？表 VII 评估了 START 在路牌提示早于实际路口被观察到的情况下，能否执行转弯。在此设置中，当前路牌提示在距离路口一定距离后被移除，迫使智能体依赖之前的提示观察和当前场景几何信息。START 在 0.5 米到 1.5 米的延迟距离范围内，性能基本保持不变，SR 约为 0.49 ~ 0.50，CSFL 保持在 0.57。在 2.0 米处，性能开始下降，此时提示需要保留更长时间，且更大的轨迹不确定性可能累积。这些结果表明，START 能够通过时序记忆和空间 grounding 处理中等程度的延迟转弯情况，而更长的提示 - 动作延迟仍具有挑战性。

不同输入模态的影响是什么？表 VIII 报告了 RGB 和深度观测对 START 的影响。与 RGB-D 输入相比，仅深度变体实现了最佳性能，将 SR 从 0.49 提升至 0.56，将 CSFL 从 0.57 提升至 0.62。这表明在当前 LSI-Dataset 中，几何信息为局部可通行性和路线结构提供了强有力的线索。仅 RGB 变体表现较差，表明仅凭外观信息在这些室内场景中对于闭环控制不够可靠。尽管如此，RGB 在完整的现实世界系统中对于上游标志解析和线索提取仍然是必要的。这些结果也凸显了当前基准的一个局限性：策略可能过度依赖几何结构，未来的工作应纳入更丰富的语义监督，并在几何信息本身不足、视觉多样性更高的环境中进行测试。

VI. 结论与局限性

本文提出了 SignNav，一种面向 LSI 环境的由方向性标志线索引导的视觉导航任务。我们构建了 LSI-Dataset，其中包含程序化注入的标志、多样的标志外观以及两种线索表示：视觉线索裁剪和结构化方向类别。我们还提出了 START，一种时空感知的 Transformer，该模型将目标相关的方向性线索锚定到第一人称观察中，并利用时间记忆来处理稀疏或延迟的标志。在 LSI-Dataset 上的实验、线索表示消融研究，

sparse-cue evaluations, and real-world robot trials demonstrate the effectiveness of START for downstream signage-cue conditioned navigation.

While these results show the promise of directional signage cues for LSI navigation, they also point to several limitations of the current formulation. This work focuses on parsed directional cues rather than full signage understanding, such as OCR, compound sign parsing, or open-ended visual sign grounding. The current cue and action spaces cover only basic directional behaviors and do not model richer public-space instructions such as stairs, escalators, or multi-step compound signs. The benchmark also assumes mostly static environments and does not address dynamic crowds or robust backtracking after wrong turns. Future work will extend SignNav toward richer signage semantics, more complex navigation behaviors, and stronger recovery mechanisms.

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